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Clustering Longitudinal Data: A Review of Methods and Software Packages

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Summary

Clustering of longitudinal data is becoming increasingly popular in many fields such as social sciences, business, environmental science, medicine and healthcare. However, it is often challenging due to the complex nature of the data, such as dependencies between observations collected over time, missingness, sparsity and non-linearity, making it difficult to identify meaningful patterns and relationships among the data. Despite the increasingly common application of cluster analysis for longitudinal data, many existing methods are still less known to researchers, and limited guidance is provided in choosing between methods and software packages. In this paper, we review several commonly used methods for clustering longitudinal data. These methods are broadly classified into three categories, namely, model-based approaches, algorithm-based approaches and functional clustering approaches. We perform a comparison among these methods and their corresponding R software packages using real-life datasets and simulated datasets under various conditions. Findings from the analyses and recommendations for using these approaches in practice are discussed.

Key words: cluster analysis; longitudinal data; model-based clustering; algorithm-based clustering; functional clustering.

1 Introduction

Clustering, a widely used data analysis technique, involves grouping individuals into clusters based on similarities in their observations. When applied to longitudinal data, clustering aims to group individuals according to the patterns and trends observed in their repeated measurements over time. Longitudinal data refer to a type of data that is collected by repeatedly measuring the same variables or attributes on the same individual over time (Diggle et al., 2002; Laird, 2022). Clustering of longitudinal data is becoming increasingly popular in many fields such as social sciences, business, environmental science, medicine and healthcare. This is because longitudinal data provide valuable insights into distinct patterns of change over time, predict future outcomes and inform decision-making. For example, the clustering technique was applied to understand employment patterns in social sciences (Huang et al., 2011), customer behaviour patterns over time for the retailer in business (Ha et al., 2002), pollution patterns of organic micropollutants in a river in environmental science (Beckers et al., 2020). In particular, in medicine and healthcare, clustering can help identify different subgroups of patients based on their clinical characteristics or disease progression patterns, which may reflect disease phenotypes

(Figure 1). These phenotypes could facilitate early detection and targeted intervention, leading to improving treatment outcomes and reducing healthcare costs. Examples include identifying infant body mass index growth patterns (Ali et al., 2021), Parkinson's disease developmental trajectories (Birkenbihl et al., 2023) and hospital healthcare flows (Pinaire et al., 2021). See Szczesniak et al. (2017), Toro-Domínguez et al. (2018) and Poulakis et al. (2022) for other examples.

Clustering of longitudinal data is often challenging due to the complex nature of the data, such as dependencies between observations over time, missingness, sparsity and non-linearity, making it difficult to identify meaningful patterns and relationships among the data. In particular, four different types of longitudinal data are commonly seen in practice (Figure 2). These include data collected at equal-spaced time grids and the number of measurements is identical

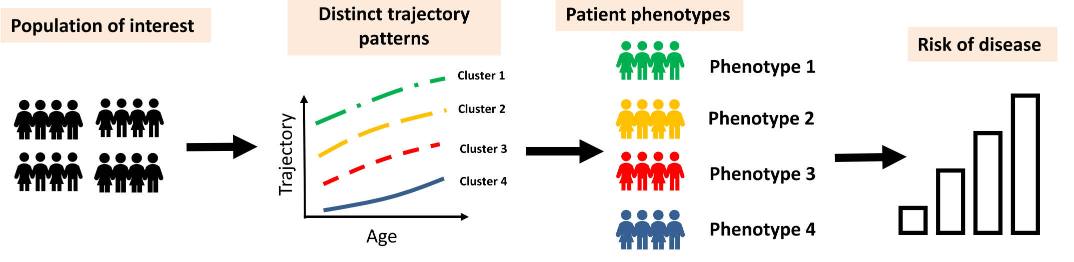

Figure 1. *Schematic diagram for clustering of longitudinal data to identify patient phenotypes.*

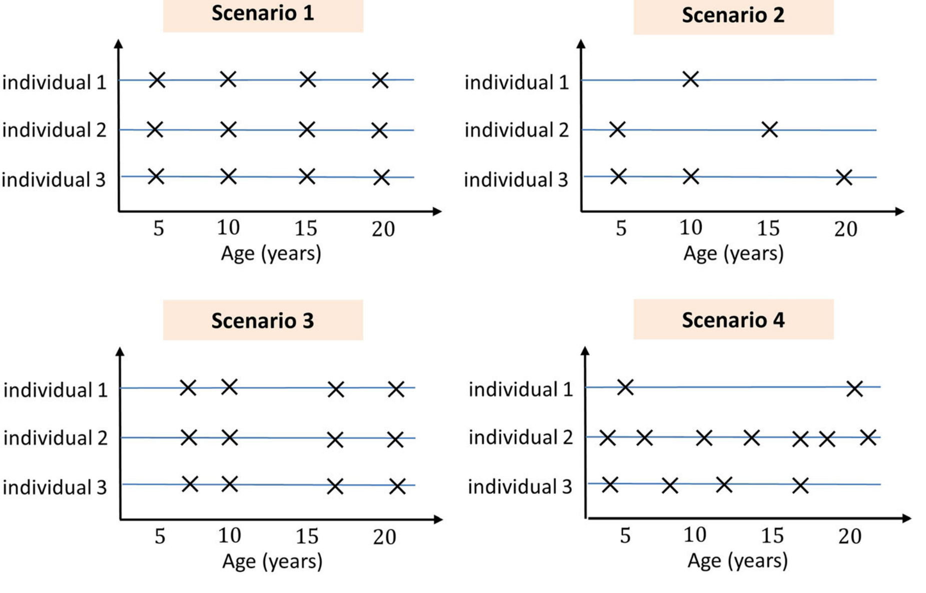

Figure 2. *Examples of commonly seen longitudinal data structures. Scenario 1 (equal-time spaced and balanced data): data are collected at equally spaced time grids and the number of measurements is identical across individuals. Scenario 2 (equal-time spaced and unbalanced data): data are collected at equally spaced time grids and the number of measurements differs across individuals due to missingness. Scenario 3 (unequal-time spaced and balanced data): data are collected at unequally spaced time grids and the number of measurements is identical across individuals. Scenario 4 (unequal-time spaced and unbalanced data): data are collected at unequally spaced time grids and the number of measurements differs across individuals. Note: The cross sign indicates the measurement is recorded at that given time point.*

across individuals (Scenario 1), data collected at equally spaced time grids but the number of measurements differs across individuals (also known as unbalanced longitudinal data) due to missingness (Scenario 2), data collected at unequally spaced time grids (also known as irregular longitudinal data) and the number of measurement is identical across individuals (Scenario 3), and data collected at unequally spaced time grids and the number of measurements differs across individuals (Scenario 4). Analytical methods may differ depending on the types of data at hand.

Various clustering approaches have been developed to handle longitudinal data. These approaches can be broadly categorised into three types: model-based, algorithm-based and functional clustering approaches. Specifically, model-based clustering assumes that the data are generated by a mixture of underlying probability distributions, with each mixture component corresponding to a cluster. The finite mixture model (McLachlan & Peel, 2004) is a popular model-based clustering method. Example methods that fall under this category include group-based trajectory models (also known as latent class growth analysis) (Nagin, 1999; Nagin & Odgers, 2010), which assume individual trajectories are identical within each cluster, and latent class mixed effect models (also known as growth mixture models) (Dunson et al., 2008; Muthén & Shedden, 1999; Proust-Lima et al., 2017), which allow individual trajectories vary around the mean trajectory of each cluster modelled by random effects. Models for clustering a special type of longitudinal data, such as skew data (Juárez & Steel, 2010; McNicholas & Subedi, 2012), incomplete data (Allen et al., 2021; Shaikh et al., 2010), non-linear data (Ding et al., 2021; Lu & Lou, 2019), multi-source data (Lu, Subbarao, & Lou, 2023) and data with high dimensional covariates (Lu & Lou, 2021) have also been developed. See Herle et al. (2020), McLachlan et al. (2019), Melnykov & Maitra (2010) and van der Nest et al. (2020) for reviews on model-based clustering.

While model-based approaches require strong assumptions about the underlying probability distribution of the clustering variables, algorithm-based approaches, on the other hand, do not make assumptions about the underlying data distribution. Furthermore, unlike model-based approaches which capture the within-subject correlation between repeated measurements through random effects, algorithm-based approaches do not explicitly take into account this correlation. Instead, algorithm-based approaches group similar observations based on some distance metric or similarity measure. Due to this reason, they are also known as non-parametric approaches. The computational speed of these approaches is generally faster and can handle small datasets that do not meet distributional assumptions. Methods that fall under this category include K-means clustering (Genolini et al., 2016; Genolini & Falissard, 2010), hierarchical clustering (Zhou et al., 2023) and correlation-based clustering (Pinto da Costa et al., 2021).

Functional data analysis (FDA) is an alternative approach to longitudinal data analysis (LDA). In many applications, FDA and LDA are connected with common goals of understanding the developmental patterns of certain characteristics measured repeatedly over time (Wang et al., 2016). While the connection between FDA and LDA has been explored in the literature (Mu, 2008; Wang et al., 2016; Zhao et al., 2004), the application of FDA to longitudinal study is still relatively novel, particularly in medical research (Ullah & Finch, 2013). Unlike LDA in which data are viewed as individual data points, in FDA the data are viewed as functions, and therefore, the observations are assumed to come from an infinite dimensional space. The within-subject correlation is captured by modelling the covariance function of the stochastic process. One important aspect of FDA is the choice of a suitable representation for the data. This can involve selecting an appropriate basis function (e.g. spline basis) or using functional principal component analysis (FPCA) to identify the dominant modes of variation in the data. In particular, modelling approaches may differ substantially in FDA for sparse and dense data, given the distinct estimation procedures and asymptotic properties (Zhang & Wang, 2016).

While the formal definitions of dense and sparse data are still lacking, for dense data (with a large number of observations), the observations from each individual are often pre-smoothed to remove noise and to restructure the random curve of that individual, before subsequent analysis (e.g. clustering) (Ramsay & Silverman, 2005; Zhang & Chen, 2007). For sparse data (with a small number of observations), such strategies are difficult due to the limit of data points. Instead, data from all individuals are pooled to borrow information from each other before subsequent analysis (James et al., 2000; Paul & Peng, 2009; Yao et al., 2005; Zhang & Wang, 2016).

There have been extensive methods developed for clustering under the framework of FDA (also known as curve clustering). Jacques and Preda (2014a) divided the functional clustering approaches into four categories. The first category (raw data methods) refers to approaches directly analysing the data points of the curves. In such a case, longitudinal data are treated as multivariate data and a finite mixture model is applied for clustering purposes. This category of methods coincides with the model-based clustering approach under LDA mentioned above. Further discussion of methods under this category can be found in Bouveyron et al. (2019) and Bouveyron & Brunet-Saumard (2014). The second category (filtering methods) refers to approaches that first approximate the curves into a finite basis of functions as a way of dimension reduction and subsequently perform clustering on the basis of expansion coefficients. For example, curve approximation is performed via B-splines (Abraham et al., 2003) or eigenfunctions (Peng & Müller, 2008) followed by a K-means clustering. The third category (adaptive methods) refers to methods performing simultaneously dimensional reduction of the curves and clustering. Examples include model-based clustering of basis expansion coefficients (Giacofci et al., 2013; Guo et al., 2022; Heard et al., 2006; James & Sugar, 2003; Ray & Mallick, 2006; Samé et al., 2011) or functional principal components analysis (FPCA) scores (Bouveyron & Jacques, 2011; Chiou & Li, 2007; Jacques & Preda, 2014b). The last category (distance-based methods) refers to approaches using clustering algorithms based on specific distances for functional data. Examples include the K-means (Cuesta-Albertos & Fraiman, 2007; Tarpey, 2007; Tokushige et al., 2007) and hierarchical clustering (Ferraty & Vieu, 2006). See Jacques & Preda, (2014a) and Zhang & Parnell (2023) for reviews on functional clustering.

Despite the increasingly common application of cluster analysis for longitudinal data, many of these existing methods are still less known to researchers, and limited guidance is provided in choosing between methods and software packages. Moreover, much of the review literature has been focused on a single category of methods. Because of this reason, the connection and comparison between methods for LDA and FDA in the context of cluster analysis have not been fully explored. To this end, this paper aims to review the recent advances in methods and software packages for clustering longitudinal data. Given the large number of literature and methods related to this topic, we do not attempt to provide a comprehensive review of this rapidly evolving field. Instead, we focus on selected methods for clustering continuous longitudinal data that can be implemented using the R software. Comparisons between these approaches based on real datasets and simulated datasets are presented and discussed.

The rest of this paper is organised as follows: in Section 2, we provide a brief review of the methodological backgrounds of the analytical approaches for clustering longitudinal data, which are broadly categorised into model-based, algorithm-based and functional clustering approaches. In Section 3, we summarise methods for determining the number of clusters for each category of approach. In Section 4, we summarise the available R software packages for performing clustering for longitudinal data. In Section 5, we provide case studies based on four real-life data with distinct structures to demonstrate and compare the performance of selected approaches. In Section 6, we perform a simulation study to compare these approaches under various conditions. Finally, in Section 7, we discuss our findings and offer some recommendations for using these approaches in practice.

2 Methods

In this section, we provide an overview of the methodological backgrounds for the three categories of approaches, namely, the model-based, algorithm-based and functional clustering approaches. The underlying assumption is that the study population is heterogeneous and is composed of K subgroups (i.e. clusters).

2.1 Model-Based Approaches

Let $\boldsymbol{y}_i = \left(y_{i1}, \ldots, y_{in_i}\right)^\top$ denote a vector of continuous observations or measurements over time for an individual i, where $i = 1, \ldots, N$ and n_i denotes the number of observations for individual i, which indicates the number of observations could differ between individuals. Also, let y_{ij} denote the j-th observation for the i-th individual, for $j = 1, \ldots, n_i$ and $i = 1, \ldots, N$. Under the framework of a finite mixture model (McLachlan & Peel, 2004), the density function of the observed data, denoted as $P(\boldsymbol{y}_i)$, can be decomposed into K components,

$$P(\boldsymbol{y}_i) = \sum_{k=1}^{K} \pi_k P_k(\boldsymbol{y}_i) \tag{1}$$

where π_k are mixing proportions or weights, for $k = 1, \ldots, K$, $0 \leq \pi_k \leq 1$ and $\sum_{k=1}^{K} = 1$. Let C_i denote the cluster membership for individual i, where $C_i = 1, \ldots, K$, and then the mixing proportions can be further written as $\pi_k = P(C_i = k)$ and can be interpreted as the probability of individual i belongs to cluster k, where $k = 1, \ldots, K$. Also, $P_k(\boldsymbol{y}_i)$ is the cluster-specific probability or density function for cluster k. It is not uncommon to assume that P_k has a parametric form characterised by a set of parameters $\boldsymbol{\theta}_k$; therefore, (1) can be written as

$$P(\boldsymbol{y}_i|\boldsymbol{\theta}) = \sum_{k=1}^{K} \pi_k P_k(\boldsymbol{y}_i|\boldsymbol{\theta}_k) \tag{2}$$

where $\boldsymbol{\theta} = (\boldsymbol{\pi}, \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K)$.

Alternatively, the mixture model can be viewed as a missing data model. Introducing a dummy latent variable $\boldsymbol{c}_i = (c_{i1}, \ldots, c_{iK})$, where $c_{ik} = 1$ if $\boldsymbol{y}_i$ belongs to cluster k, and $c_{ik} = 0$ otherwise. In addition, $\boldsymbol{c}_i$ is assumed to follow a multinomial distribution consisting of one draw on K categories with probability $\pi_1, \ldots, \pi_K$, that is, $\boldsymbol{c}_i \sim \text{Multinomial}(1; \boldsymbol{\pi})$. Under this model, the log-likelihood considered both $\boldsymbol{y}_i$ and $\boldsymbol{c}_i$ as data (also known as complete-data log-likelihood) is

$$l(\boldsymbol{\theta}|\boldsymbol{y}_i, \boldsymbol{c}_i) = \sum_{k=1}^{K} c_{ik}(\log\pi_k + \log P_k(\boldsymbol{y}_i|\boldsymbol{\theta}_k)) \tag{3}$$

To estimate the mixture model, the maximum likelihood method via the Expectation–Maximisation (EM) algorithm is a popular approach (Dempster et al., 1977). Briefly, the EM algorithm consists of two main steps: the expectation step (E-step) and the maximisation step (M-step). In the E-step, the algorithm estimates the probability distribution of the latent variables ($\boldsymbol{c}_i$ given the observed data and the current estimates of the model parameters. This step involves computing the expected value of the complete-data log-likelihood function with respect to the conditional probability distribution of the latent variables. In the M-step, the algorithm updates the estimates of the model parameters based on the expected values computed in the E-step. This step involves finding the parameters that maximise the expected complete-data log-likelihood function. The algorithm alternates between the E-step and M-step until the parameter estimates no longer change or the maximum number of iterations is reached.

Bayesian inference provides an alternative tool to estimate the mixture model. Within the Bayesian framework, the Markov chain Monte Carlo (MCMC) method refers to a class of algorithms used for sampling from the probability distributions of parameters of interest whereas the Gibbs sampler is the most popular MCMC algorithm used in finite mixture models (Chib, 1995; Diebolt & Robert, 1994; Escobar & West, 1995). The Gibbs sampling algorithm iteratively samples from the posterior distribution of the mixture model parameters and the latent variables. In each iteration, the algorithm updates the parameters of each cluster based on the assigned observations and then updates the cluster assignments based on the posterior distribution. The process continues until convergence is reached. Despite their differences in application, the EM algorithm and Gibbs sampling algorithm can be connected in the context of mixture models. For example, the E-step of the EM algorithm can be seen as a special case of Gibbs sampling where only the latent variables are updated, while the M-step of the EM algorithm can be seen as a closed-form solution to the parameter updates in the Gibbs sampling algorithm.

Finite mixture models for longitudinal data are well developed. In the following subsection, we provide an overview of two classes of models that have been widely used, namely, the group-based trajectory model (GBTM) and the latent class mixed effect model (LCMM). Both GBTM and LCMM are a variant of the finite mixture model.

2.1.1 Group-Based Trajectory Model

GBTM (also known as latent class growth analysis) models the population-level distribution of trajectories and estimates the probability of each individual belonging to each group based on their observed data (Nagin, 1999). The key idea of GBTM is to model the individual trajectories as a function of time, and then group individuals based on similarities in their trajectories. The model assumes that the trajectory for each individual within a group follows a similar pattern, but the parameters (e.g. intercept and slope) may differ between groups. The goal is to identify the number and pattern of the trajectory groups that best explain the observed data. Mathematically, the GBTM can be written as

$$P(\boldsymbol{y}_i|\boldsymbol{x}_i,\, \boldsymbol{\theta}) = \sum_{k=1}^{K} \pi_k P(\boldsymbol{y}_i|\boldsymbol{x}_i, \boldsymbol{\theta}_k) \tag{4}$$

where $\boldsymbol{x}_i = (x_{i1}, \ldots, x_{in_i})^\top$ denote the vector of time where $\boldsymbol{y}_i$ is recorded. Furthermore, GBTM assumes that the repeated measurements are independent conditional on the latent class membership k, which is known as conditional independence assumption. With this assumption, (4) can be further expressed as

$$P(\boldsymbol{y}_i|\boldsymbol{x}_i,\, \boldsymbol{\theta}) = \sum_{k=1}^{K} \pi_k \prod_{j=1}^{n_i} P\left(y_{ij}|x_{ij}, \boldsymbol{\theta}_k\right) \tag{5}$$

The cluster-specific density function $P\left(y_{ij}|x_{ij}, \boldsymbol{\theta}_k\right)$ is typically assumed to be a normal distribution with its mean modelling by a polynomial function of time or age. For example, a linear function can be used, that is, $y_{ij}\,|\,x_{ij},\, \beta_{0k},\, \beta_{1k},\, \sigma_k^2 \sim N\left(\beta_{0k} + \beta_{01}x_{ij}, \sigma_k^2\right)$, where $\boldsymbol{\theta}_k = \left(\beta_{0k}, \beta_{1k}, \sigma_k^2\right)$, β_{0k} and β_{1k} are coefficients of the polynomial function, and σ_k^2 is the residual variance for cluster k, where $k = 1,\, \ldots,\, K$.

GBTM is often implemented using a maximum likelihood estimation via the EM algorithm, where the likelihood function for an individual i has the same form as (5). The assignment to a cluster is based on the calculation of the posterior cluster membership probabilities based on estimated parameters, denoted as $P\left(C_i = k|\boldsymbol{y}_i, \boldsymbol{x}_i, \widehat{\boldsymbol{\theta}}\right)$, which is estimated by

$$P\left(C_i = k|\boldsymbol{y}_i, \boldsymbol{x}_i, \widehat{\boldsymbol{\theta}}\right) = \frac{P\left(\boldsymbol{y}_i|\boldsymbol{x}_i, \widehat{\boldsymbol{\theta}}_k\right)\widehat{\pi}_k}{\sum_{l=1}^{K} P\left(\boldsymbol{y}_i|\boldsymbol{x}_i, \widehat{\boldsymbol{\theta}}_l\right)\widehat{\pi}_l} \tag{6}$$

where $C_i = 1, \ldots, K$ denotes the cluster membership for an individual i, $\widehat{\boldsymbol{\theta}}_k$ is the estimated model parameters and $\widehat{\pi}_k$ is the estimated proportion of the population in cluster k. Individuals are then assigned to the cluster to which their posterior membership probability is the largest. The posterior cluster membership probabilities can be used to assess the goodness-of-fit of the model (e.g. for the selection of the number of clusters) and the discrimination of the clusters.

2.1.2 Latent Class Mixed Effect Model

Unlike GBTM which assumes individual trajectory is identical within each cluster k, LCMM (also known as latent growth mixture modelling) allows for the estimation of within-class variation in growth trajectories (Muthén & Shedden, 1999; Proust-Lima et al., 2017). LCMM can be viewed as an extension of GBTM and uses random effects to capture deviations of individual trajectories within each cluster k. The random effects serve the purpose of capturing the correlation between repeated measurements of an individual. Specifically, in LCMM, the cluster-specific probability density function $P(\boldsymbol{y}_i|\boldsymbol{x}_i, \boldsymbol{\theta}_k)$ in (4) can be modelled by a mixed effect model, that is,

$$\boldsymbol{y}_i|\boldsymbol{x}_i,\, \boldsymbol{\theta}_k \sim N\left(\boldsymbol{x}_i^\top \boldsymbol{\beta}_k + \boldsymbol{Z}_i^\top \boldsymbol{b}_k,\, \boldsymbol{\Lambda}_k\right) \tag{7}$$

where $\boldsymbol{x}_i^\top$ and $\boldsymbol{Z}_i$ denote the design matrix for the fixed effect and random effect covariates, and $\boldsymbol{\theta}_k = (\boldsymbol{\beta}_k, \boldsymbol{b}_k)$, where $\boldsymbol{\beta}_k$ and $\boldsymbol{b}_k$ are vectors of coefficients for fixed effect and random effect of cluster k, respectively. Λ_k is the variance–covariance matrix. For long series with a large number of observations, Λ_k can be constructed to capture the serial correlation, for example, using an autoregressive structure with an order of 1. Typically, $\boldsymbol{b}_k$ is assumed to have a multivariate normal distribution with a mean vector of $\mathbf{0}$, that is, $\boldsymbol{b}_k \sim N(0,\, \boldsymbol{\Sigma}_k)$; however, this assumption can be relaxed (e.g. using multivariate t distribution) to allow for more flexibility.

While LCMM specified in (7) assumes both fixed effect $\boldsymbol{\beta}_k$ and random effect $\boldsymbol{b}_k$ differ between cluster k, a variant of LCMM assumes that the fixed effect is shared between clusters and only the random effect is different between clusters (Komárek & Komárková, 2013). In such a case, $\boldsymbol{\beta}_k$ in (7) is assumed to be common across clusters, that is, $\boldsymbol{\beta}_1 = \ldots = \boldsymbol{\beta}_K = \boldsymbol{\beta}$. The EM-type algorithm (Muthén & Shedden, 1999), the Newton–Raphson-type algorithm (Proust & Jacqmin-Gadda, 2005; Proust-Lima et al., 2017) and the MCMC algorithm (Komárek & Komárková, 2013) have been developed to estimate the LCMM model. As with GBTM, LCMM assigns an individual to a cluster based on the posterior cluster membership probability defined in (6).

2.2 Algorithm-Based Approaches

Algorithm-based approaches tend to partition a group of individuals into K distinct, non-overlapping clusters. Commonly used algorithm-based approaches, such as K-means clustering (Genolini et al., 2016; Genolini & Falissard, 2010) and hierarchical clustering (Van Den Bergh & Vermunt, 2018; Zhou et al., 2023) have been extended to longitudinal data.

2.2.1 K-Means Clustering

K-means clustering is a popular unsupervised learning algorithm used for grouping data points into K number of clusters based on their similarities. The algorithm starts by randomly selecting K centroids (cluster centres) from the dataset. Then, each individual is assigned to the nearest centroid based on the pre-defined distance metric. The algorithm can also be considered an EM algorithm. During the E-step, the centre of each cluster is determined, and then during the M-step, each individual is assigned to its 'nearest cluster'. The alternation of the two steps is repeated until no further changes occur in the clusters.

In a longitudinal data setting, 'cluster centres' are defined as the mean trajectory of each cluster, which is the mean of trajectories over all individuals who belong to that cluster. The K-means clustering does not explicitly take into account the correlation within individuals. Instead, the key is to define the distance. To appropriately define the distance between two individuals based on longitudinal observations, the K-means method requires complete and balanced longitudinal data (Figure 2 Scenarios 1 and 3), that is, $n_i = n$, for $i = 1, \ldots, N$. For example, for two individuals i and i', popular distance metrics include the Euclidean distance, which is defined as $D(\boldsymbol{y}_i, \boldsymbol{y}_{i'}) = \sqrt{\sum_{l=1}^{n} (y_{il} - y_{i'l})^2}$ and the Manhattan distance, which is defined as $D(\boldsymbol{y}_i, \boldsymbol{y}_{i'}) = \sum_{l=1}^{n} |y_{il} - y_{i'l}|$. When missing data are present, imputation methods could be used to fill in the missingness.

2.2.2 Hierarchical Clustering

Hierarchical clustering is another popular unsupervised machine learning algorithm. The algorithm builds a hierarchy of clusters that can be represented as a dendrogram, which is a tree-like diagram showing the nested clusters at different levels of similarity (Murtagh & Contreras, 2012, 2017). There are two main types of hierarchical clustering: agglomerative and divisive. Agglomerative clustering starts by treating each individual as a separate cluster and then iteratively merges the closest clusters until a single cluster is formed. Divisive clustering, on the other hand, starts with a single cluster containing all the individuals and then recursively divides it into smaller clusters until each cluster contains a single individual.

Recently, Zhou et al. (2023) proposed a flexible hierarchical clustering method that can be applied to different types of longitudinal data, including Scenarios from 1 to 4 (Figure 2). This method uses B-splines De Boor et al. (1978) with p basis functions to approximate the individual trajectory. Specifically, let $\boldsymbol{x}_{ij} = \left[B_1\left(t_{ij}\right), \ldots, B_p\left(t_{ij}\right)\right)]$ denote the row vector of the B-splines basis function at t_{ij}. Generally, based on all individuals, the model can be written as $y_{ij} = \boldsymbol{x}_{ij}\boldsymbol{\beta}_i + \epsilon_{ij}$, where $\boldsymbol{\beta}_i = \left(\beta_{i1}, \ldots, \beta_{ip}\right)^\top$ are the B-spline coefficients. No distributional assumptions are imposed on these coefficients and they are estimated by the ordinary least-squares method. ϵ_{ij} is the random noise for individual i at time t_{ij}, where $E\left(\epsilon_{ij}\right) = 0$ and $Var\left(\epsilon_{ij}\right) = \sigma^2$. The σ^2 can be estimated by fitting the model with all the data. Similar to K-means clustering, hierarchical clustering does not explicitly model the within-individual correlation.

The hierarchical clustering approach uses a metric to quantify the cost of combining two sub-clusters in order to determine the appropriateness of clustering. Denote the observed data from cluster k as $O_k := \left\{ \left(\boldsymbol{x}_{ij}, y_{ij}\right) | \text{for all } i \text{ for which } C_i = k, j = 1, \ldots., n_i \right\}$, for $k = 1, \ldots, K$.

Also, let $O_{u,v}$ be the union of O_u and O_v, and then $O_{u,v} = \{O_u, O_v\}$. A metric of dissimilarity between two subsets O_u and O_v can be defined as

$$D(O_u, O_v) = \frac{\left(\mathrm{SSR}\left(O_{u,v}\right) - \mathrm{SSR}(O_u) - \mathrm{SSR}(O_v)\right)/p}{(\mathrm{SSR}(O_u) - \mathrm{SSR}(O_v))/(n_u + n_v - 2p)} \tag{8}$$

where $\mathrm{SSR}(O_k)$ is the sum of squared residuals (SSR) and is calculated as $\mathrm{SSR}(O_k) = \sum_{i=1}^{N}\sum_{j=1}^{n_i} c_{ik}\left(y_{ij} - \widehat{y}_{ij}\right)^2$. p is the dimension of the B-spline basis. Also, $n_u = \sum_{i=1}^{N} c_{iu}n_i$, $n_v = \sum_{i=1}^{N} c_{iv}n_i$ and $\widehat{y}_{ij} = \widehat{f}_k\left(t_{ij}\right) = \boldsymbol{x}_{ij}\widehat{\boldsymbol{\beta}}_k$, where $\widehat{\boldsymbol{\beta}}_k$ is the ordinary least-squares estimate for $\boldsymbol{\beta}_k$, based on the data from all individuals belonging to cluster k. For example, a larger $D(O_u, O_v)$ value suggests a stronger dissimilarity between O_u and O_v and thus greater evidence against merging the two subsets.

A greedy search algorithm was developed (Zhou et al., 2023), which is an agglomerative (i.e. bottom-up) approach in which the size of the clusters increases (and the number of clusters decreases) as the algorithm proceeds. The algorithm begins with each individual as a single cluster; then fits the B-spline model and calculates SSRs. Subsequently, the algorithm conducts a greedy search to merge two subclusters with the minimal cost metric $D(O_u, O_v)$ into one cluster and then updates SSRs after each merge (bottom-up merging process). This process is repeated hierarchically to the top level where only one cluster is left.

2.3 Functional Clustering Appraoches

In FDA, the observations are assumed to belong to an infinite dimensional space. Therefore, a common strategy for functional clustering approaches is to first find representative curve patterns through smoothing and dimension reduction techniques (Wang et al., 2016), based on data from all individuals. Once the functional data have been represented and dimensionally reduced, traditional clustering techniques can be applied to group similar functions together. These techniques may include K-means clustering, hierarchical clustering or model-based clustering methods. The choice of clustering method depends on factors such as the distributional assumptions of the data and the desired properties of the clusters (e.g. shape, size and number of clusters). Given that it is a critical step in functional clustering approaches, this subsection provides an overview of common methods for finding representative curve patterns through smoothing and dimension reduction techniques.

Let $G(\cdot)$ define a stochastic process on an interval $I = [0, T]$, with mean function $\mu(t) = E(G(t))$ and the covariance function $\boldsymbol{\Sigma}(s, t) = \mathrm{cov}(G(s), G(t))$. The realisation of this process for an individual i is $G_i = G_i(\cdot)$, for $i = 1, \ldots, N$. To allow the number of measurements to differ between individuals, we let the observations for an individual i to be $\boldsymbol{G}_i = \left(G_{i1}, \ldots, G_{in_i}\right)$, where $G_{ij} = G_i\left(t_{ij}\right)$ and $j = 1, \ldots, n_i$. In the special case when observations are collected at fixed time points, and then we have $t_{ij} = t_{\cdot j}$ and $n_i = n$ for all i (Figure 2, Scenarios 1 and 3). In case when the observations are collected at an equally spaced time interval, and then we have $t_{\cdot j} - t_{\cdot j-1} = t_{\cdot j+1} - t_{\cdot j}$ for all j (Figure 2, Scenario 1). Also, let ϵ_{ij} denote the random noise (also known as measurement errors), with $E\left(\epsilon_{ij}\right) = 0$ and $\mathrm{Var}\left(\epsilon_{ij}\right) = \sigma_{ij}^2$. The ϵ_{ij} is assumed to be independent across i and j. Therefore, the actual observed value can be written as $y_{ij} = G_{ij} + \epsilon_{ij}$.

FDA usually consists of approximating the curves into a finite basis of functions (known as the filtering step). Popular methods for this approximation include using spline basis (e.g. B-spline basis), functional principal component analysis (FPCA), wavelet and kernel smoothing. For example, let $\varphi = \left\{\varphi_1, \ldots, \varphi_p\right\}$ represents a p dimensional spline basis, then

$$y_{ij} = f_i\left(t_{ij}\right) + \epsilon_{ij} = \sum_{l=1}^{p} a_l \varphi_l\left(t_{ij}\right) + \epsilon_{ij} \tag{9}$$

where $\boldsymbol{a} = \left(a_1, \ldots, a_p\right)^\top$ is a vector of coefficients and p is the number of regression coefficients. The overall trends of y_{ij} can be approximated through the B-spline basis by letting $\varphi_l\left(t_{ij}\right) = B_l\left(t_{ij}\right)$; see Abraham et al. (2003), Luan and Li (2003) and Song et al. (2007) for example. To induce clustering, the basis expansion coefficients $\boldsymbol{a}$ are assumed to be random and follow a mixture of Gaussian distribution, with means μ_k and common variance–covariance matrix $\boldsymbol{\Sigma}$, that is, $a_l \sim N(\mu_k, \boldsymbol{\Sigma})$, for $l = 1, \ldots, p$, which allows efficiently modelling sparse functional data (James & Sugar, 2003). Alternatively, a hierarchical Bayesian model can be considered, by imposing a prior distribution on the coefficients $\boldsymbol{a}$, that is, $a_l|\tau_k \sim N(\mu, \tau_k\boldsymbol{\Sigma})$ and $\tau_k \sim \mathrm{IG}(u, v)$, where $\mathrm{IG}(u, v)$ denote an inverse-gamma distribution with parameters u and v. For example, see Heard et al. (2006) and Ray et al. (2014).

Apart from the spline basis, FPCA is a widely used tool for identifying the major source of variations in the curves and projecting infinite-dimensional curves into finite-dimensional vectors (Ramsay & Silverman, 2005). Let $y_i(t)$ be the functional data of the i-th individual, with $t \in T$ where T is the bounded time range. Based on the Karhunen-Loeve theorem, $y_i(t)$ can be written as $y_i(t) = f_i(t) + \epsilon_i(t) = \mu(t) + \sum_{q=1}^{\infty}\xi_{iq}\phi_q(t) + \epsilon_i(t)$, where $E(\epsilon_i(t)) = 0$ and $Var(\epsilon_i(t)) = \sigma^2$. The $\phi_q(t)$ is the q^{th} functional principal component (FPC) and $\xi_{iq} = \int_T (f_i(t) - \mu(t))\phi_q(t))dt$ is the associated FPC score. For longitudinal data, it is common that we only observed (very few) discrete observations y_{ij} of sample curve $y_i(t)$ at a finite set of time points t_{ij} (Figure 2; Scenario 4). In such cases, conditional expectation (PACE) method (Yao et al., 2005) is commonly used; then y_{ij} can be expressed as

$$y_{ij} = f_i\left(t_{ij}\right) + \epsilon_{ij} = \mu\left(t_{ij}\right) + \sum_{q=1}^{\infty} \xi_{iq}\phi_q\left(t_{ij}\right) + \epsilon_{ij} \tag{10}$$

Typically, the function $f_i\left(t_{ij}\right)$ is well approximated by the first few (e.g. Q) leading FPCs and FPC scores, which leads to

$$y_{ij} = \mu\left(t_{ij}\right) + \sum_{q=1}^{Q} \xi_{iq}\phi_q\left(t_{ij}\right) + \epsilon_{ij} \tag{11}$$

To estimate the mean function $\mu\left(t_{ij}\right)$, two difference scenarios can be considered. When the observations are available on a regular grid (data can be treated as multivariate data) (Figure 2; Scenarios 1 and 3), one can take the average at each time point t_{ij}, that is, $\widehat{\mu}\left(t_{ij}\right) = \frac{1}{n}\sum_{i=1}^{n} y_{ij}$. When the observations are sparse and measured on an irregular grid (Figure 2 Scenarios 2 and 4), $\mu\left(t_{ij}\right)$ is obtained by smoothing the data from all observations based on the local linear smoother method (Fan, 1996). Smoothing techniques are then used to estimate the variance and covariance function (Wang et al., 2016; Yao et al., 2005). The FPC score of the i-th curve $y_i(t)$ on the q-th FPC can be obtained by the conditional expectation $\widehat{\xi}_{iq} = E\left(\xi_{iq}|\boldsymbol{y}_i\right) = \lambda_q\widehat{\boldsymbol{\phi}}_q^\top\widehat{\boldsymbol{\Sigma}}_{\boldsymbol{y}_i}^{-1}(\boldsymbol{y}_i - \widehat{\boldsymbol{\mu}})$, where $\lambda_1 > \lambda_2 > \ldots > 0$. $\widehat{\boldsymbol{\phi}}_q$ and $\widehat{\boldsymbol{\mu}}$ are vectors by evaluating $\widehat{\boldsymbol{\phi}}_q(t)$

and $\hat{\mu}(t)$ at time point t_{ij} for $j = 1, \ldots, n_i$, $\boldsymbol{y}_i = (y_{i1}, \ldots, y_{in_i})$, $\hat{\boldsymbol{\Sigma}}_{\boldsymbol{y}_i} = \widetilde{\boldsymbol{G}} + \sigma^2 \boldsymbol{I}_{n_i}$, and the matrix $\widetilde{\boldsymbol{G}}$ is obtained by evaluating $\widehat{G}(s, t)$ at the grid points t_{ij} for $j = 1, \ldots, n_i$. The optimal number of FPCs Q can be determined by many methods (Yao et al., 2005), such as AIC, BIC, fraction-of-variance-explained (FVE) method or cross-validation (CV) based on leave-one-subject-out prediction error. Following this procedure, we can obtain a set of FPC scores $(\boldsymbol{\xi_1}, \ldots, \boldsymbol{\xi_N})$, where $\boldsymbol{\xi_i} = (\xi_{i1}, \ldots, \xi_{iQ})^\top$, for $i = 1, \ldots, N$, is a Q-dimensional vector of FPC scores for individual i. Subsequently, traditional clustering techniques such as the K-means method (Hartigan & Wong, 1979) can be applied for grouping individuals based on their FPC scores. See Chiou and Li (2007) and Dong et al. (2017) for examples.

In addition to the two-stage approach based on FPCA, a functional adaptive density peak (FADP) clustering approach (Ren et al., 2023) was proposed recently. This approach was adapted from a clustering algorithm by search and finding of density peak (Rodriguez & Laio, 2014), and it is based on two assumptions: (i) a cluster centre is surrounded by data points whose local densities (i.e. distribution of data points within a specific region) (Wang & Xu, 2017) are lower than that of the centre and (ii) cluster centres are far enough apart from any points with a higher local density. Two quantities (known as density-distance pairs) play key roles in this approach, namely, the local density of the data $\widetilde{f}(\mathbf{y}_i)$ and the minimum distance $\widetilde{\delta}_i$ between the i-th data point and any other points with higher density. Ren et al. (2023) proposed two methods to calculate the density-distance pairs, they are based on distance metrics of FPCs (called FADP1) and Euclidian norm (called FADP2), respectively. In contrast with other popular clustering algorithms where iteration is needed to optimise an objective function, the FADP clustering algorithm is only performed in one single step which could potentially be computationally efficient. See Jacques & Preda (2014a) and Zhang & Parnell (2023) for a summary of other functional clustering approaches.

3 Determining the number of clusters

Determining the number of clusters is one of the challenging steps of cluster analysis, and it is still an active research topic. Earlier comparative work can be found in Milligan and Cooper (1985) and Hardy (1996). Comprehensive reviews of methods to determine the number of clusters for model-based clustering approaches under the frequentist framework (Bouveyron & Jacques, 2011; McLachlan & Peel, 2004; Nagin & Odgers, 2010; Nagin & Tremblay, 2005; Nylund et al., 2007; van der Nest et al., 2020) and Bayesian framework (Celeux et al., 2006; Merkle et al., 2019; Nasserinejad et al., 2017) have been developed. Also developed are reviews of methods for algorithm-based approaches (Islam et al., 2015; Kodinariya & Makwana, 2013; Mirkin, 2011) and functional clustering approaches (Jacques & Preda, 2014a, 2014b). A general review of available indices and software packages can also be found in Charrad et al. (2014). Given the large amount of existing literature, the current section only provides a broad overview of commonly used methods to determine the number of clusters. Readers may refer to the cited literature for details.

Typically, the number of clusters K is treated as a fixed quantity albeit unknown. For model-based approaches, selecting the number of clusters can be viewed as a model selection process, and model selection criteria can be implemented to compare models. The optimal K is the one that provides the best model fit according to the chosen model selection criteria. However, the model selection criteria may be different between the frequentist approach and the Bayesian approach. For the frequentist approach, model selection criteria such as Akaike's information criterion (AIC) (Akaike, 1987) and Bayesian information criterion (BIC)

(Schwarz, 1978) are commonly used, and these indices are made available in many R packages (Table 1). In particular, the BIC is often recommended for model-based clustering (Fraley & Raftery, 1998; Magidson & Vermunt, 2004; Nylund et al., 2007), which is based on penalised forms of likelihood, with a penalty for the number of estimated parameters subtracted from the log-likelihood. In cluster analysis, the number of parameters is often proportional to the number of clusters. Therefore, the BIC penalises models with a large number of clusters, which helps to avoid overfitting. Alternatively, the Lo–Mendell–Rubin (Lo et al., 2001) method and bootstrap likelihood ratio test (McLachlan & Peel, 2004) have also been developed and used in practice. Nylund et al. (2007) compared these indices for selecting the number of clusters in latent class models, factor mixture models and growth mixture models, and found that BIC outperformed the AIC, while the bootstrap likelihood ratio test provided the most consistent indicator of clusters. Of note, while consistency estimators for model selection have been developed and proven for univariate and multivariate finite mixture model (Du Roy de Chaumaray & Marbac, 2024; Keribin, 2000), few studies have examined the consistency property for clustering longitudinal data.

For models fitted under the Bayesian framework, the Bayes factor can be applied when there are more than two candidate models and can be used for comparing non-nested models. For comparing two models, the Bayes factor is defined as the ratio of the two integrated likelihoods. The Bayes factor provides a directly interpretable number that quantifies the evidence provided by the data (Kass & Raftery, 1995). However, the Bayes factor requires integration over all parameters space and therefore is computationally intensive, particularly when there are many parameters of interest. To this end, the BIC is used as an approximation of the Bayes factor, and it was shown that this approximation is reasonably good (Kass & Wasserman, 1995). An index similar to BIC that measures the model complexity in Bayesian settings is the deviance information criterion (DIC) (Spiegelhalter et al., 2002). DIC provides an overall assessment of model fit and accounts for model complexity by considering the effective number of parameters. A model with a smaller DIC value is usually preferred. DIC has been adapted to different models such as missing data models, random effect models and mixture models (Celeux et al., 2006). Apart from the above methods, other methods are also available, such as the integrated classification likelihood (ICL) (Biernacki et al., 2000), ‘no small clusters’ criterion (also known as R&M criterion) (Rousseau & Mengersen, 2011), approximate weight of evidence (Banfield & Raftery, 1993), Bayesian leave-one-out cross-validation (Geisser & Eddy, 1979), the widely applicable information criterion (also known as Watanabe–Akaike information criterion) (Watanabe, 2010) and penalised expected deviance (PED) (Plummer, 2008). In contrast to treating K as a fixed quantity, several Bayesian methods treat K as a random variable of interest, and the inference of K is considered part of the modelling process. These methods include reversible jump MCMC (Richardson & Green, 1997), birth-and-death process (Stephens, 2000), Dirichlet process mixture (Neal, 2000) and the mixture of finite mixture model (Miller & Harrison, 2018). Nasserinejad et al. (2017) performed several simulation studies and found that the R&M criterion performed consistently well under various scenarios when compared with its counterparts in determining the number of clusters; therefore, it is recommended. However, the implementation of R&M requires using Bayesian analysis software such as WinBugs and JAGS.

For algorithm-based approaches in which no likelihood is defined, criteria to measure the quality of the clustering are often used instead. Commonly used criteria include Calinski and Harabasz (CH) criterion (also known as variance ratio criterion) (Caliński & Harabasz, 1974) and Silhouette index (Rousseeuw, 1987), which selects the K that optimises the difference of within-cluster and between-cluster dissimilarity. For example, the CH creation compares the ratio of between-cluster dispersion to the within-cluster dispersion. A higher CH value indicates a

Table 1. *Summary of selected methods and packages under the categories of model-based, algorithm-based and functional clustering approaches for clustering longitudinal data.*

Methods	*R packages*	*Description*	*Criteria to determine the number of clusters*	*Major strengths*	*Major weaknesses*	*Applicable to data scenarios (Figure 2)*
Model-based approach						
Group-based trajectory clustering	gbmt (Magrini, 2022)	Finite mixture model based on EM algorithm.	AIC and BIC	Computationally efficient; can include covariates; the timing variable can be modelled explicitly	Assume repeated measurements are independent given the cluster membership	1, 2, 3, 4
Group-based trajectory clustering	flexmix (Leisch, 2004)	Finite mixture model based on EM algorithm.	AIC and BIC	Computationally efficient; can include covariates; the timing variable can be modelled explicitly	Assume repeated measurements are independent given the cluster membership	1, 3
Latent class mixed effect model	lcmm (Proust-Lima et al., 2017)	Latent class mixed effect model based on the finite mixture model and Newton–Raphson-type algorithm.	AIC and BIC	Can incorporate random effect; can model non-linear trajectory; can include covariates; the timing variable can be modelled explicitly; can account for serial correlation	Computationally intensive	1, 2, 3, 4
Latent class mixed effect model	mixAK (Komárek & Komárková, 2014)	Latent class mixed effect model based on finite mixture model and MCMC.	PED	Can incorporate random effect; can include covariates; the timing variable can be modelled explicitly; can incorporate prior knowledge	Computationally intensive	1, 2, 3, 4
Gaussian mixture model	longclust (McNicholas & Murphy, 2010)	Based on a mixture of multivariate t or Gaussian distributions with a Cholesky-decomposed covariance structure.	BIC	Computationally efficient; allows modelling the mixture of multivariate t distribution	Only applicable to balanced data and does not allow missing data; the timing variable cannot be modelled explicitly; does not allow covariates	1, 3
Gaussian mixture model	kmlcov (Subtil et al., 2017)	Clustering using the likelihood as a metric of distance; EM-type algorithm is used.	CLL, AIC and BIC	Can include covariates; the timing variable can be modelled explicitly	Computationally intensive; assume repeated measurements are independent given the cluster membership	1, 3

(*Continues*)

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Table 1 *(Continued)*

Methods	*R packages*	*Description*	*Criteria to determine the number of clusters*	*Major strengths*	*Major weaknesses*	*Applicable to data scenarios (Figure 2)*
Algorithm-based approach						
K-means clustering	kml (Genolini & Falissard, 2010)	K-means clustering based on Euclidean distance or Manathan distance.	Ray and Turi criterion, Davies and Bouldin criterion, Calinski and Harabatz criterion, AIC and BIC	Can capture non-linear trajectory with complex shape; no distributional assumptions; computationally efficient	Only applicable to balanced data and does not allow missing data; the timing variable cannot be modelled explicitly; does not allow covariates	1, 3
K-means clustering	kmlshape (Genolini et al., 2016)	K-means using shape-respecting distance.	Not available	Can capture non-linear trajectory with complex shape; no distributional assumptions; computationally efficient	Only appliable to balanced data and does not allow missing data; the timing variable cannot be modelled explicitly; does not allow covariates; no model selection criteria;	1, 3
Hierarchical clustering	clusterMLD (Zhou et al., 2023)	Hierarchical clustering based on a dissimilarity metric that quantifies the cost of merging two distinct groups of curves.	Calinski and Harabatz criterion, and gap statistics	Can capture non-linear trajectory with complex shape; computationally efficient	Does not allow covariates	1, 2, 3, 4
Functional data approach						
FPCA with K-means clustering	fdapace (Yao et al., 2005)	First estimate FPC scores then perform K-means clustering on the scores.	Calinski and Harabatz criterion, silhouette statistics, and gap statistics	Can capture non-linear trajectory with complex shape; computationally efficient	Does not allow covariates	1, 2, 3, 4
Model-based clustering for functional data	fdaMocca (Pya Arnqvist et al., 2021)	Each curve is projected onto a finite-dimensional basis and clustering is done on the resulting basis coefficients; an EM-style algorithm is used.	AIC, BIC and Shannon entropy	Can capture non-linear trajectory with complex shape; allow covariates	Computationally intensive	1, 2, 3, 4
Functional subspaces clustering	funFEM (Bouveyron & Jacques, 2015)	Cluster functional data by modelling the curves within a common and discriminative functional subspace.	AIC, BIC and ICL	Can capture non-linear trajectory with complex shape; computationally efficient	Does not allow covariates	1, 3

(*Continues*)

Table 1 *(Continued)*

Methods	*R packages*	*Description*	*Criteria to determine the number of clusters*	*Major strengths*	*Major weaknesses*	*Applicable to data scenarios (Figure 2)*
Sparse and smooth functional clustering	sasfunclust (Centofanti et al., 2023)	Finite mixture model with a likelihood function penalized with a functional adaptive pairwise fusion penalty and a roughness penalty; an EM-style algorithm is used.	Cross-validation	Can capture non-linear trajectory with complex shape	Computationally intensive	1, 2, 3, 4
Functional clustering using adaptive density peak detection	FADPclust (Ren et al., 2023)	Clustering based on adaptive density peak detection technique in which the density is estimated by functional k-nearest neighbour density estimation.	Calinski and Harabatz criterion, Silhouette statistics, Dunn statistics	Can capture non-linear trajectory with complex shape; computationally efficient	Does not allow covariates	1, 3

better separation between clusters, suggesting a more optimal clustering solution. This criterion has been adapted to longitudinal data in K-means clustering (Genolini & Falissard, 2010) and hierarchical clustering (Zhou et al., 2023). Other criteria such as gap statistics (Tibshirani et al., 2001), Ray and Turi criterion (Ray & Turi, 1999) and Davies and Bouldin criterion (Davies & Bouldin, 1979) are also used in practice. However, studies on evaluating these indices for clustering longitudinal data are still lacking.

For functional clustering approaches, AIC and BIC are often used if likelihoods are defined, for example, in model-based clustering for functional data (Bouveyron & Jacques, 2011; Giacofci et al., 2013; Ma & Zhong, 2008; Samé et al., 2011). In addition, criteria measuring the quality of the clustering are also used. For example, the within-cluster sum of squares was used in Yamamoto (2012), within-cluster sum of squares coped with gap statistics was used in Floriello and Vitelli (2017), CH index was used in Park and Ahn (2017), Silhouette index was used in Martino et al. (2019), and Kullback–Leibler information criterion was used in Guo et al. (2022). On the other hand, methods assuming K is random and imposing a prior distribution (e.g. Dirichlet process) were also developed; see Heard et al. (2006), Ray & Mallick (2006), Rodríguez et al. (2009) and Suarez & Ghosal (2016) for examples.

In summary, choosing a method for determining the number of clusters depends on the cluster analysis methods being used, the availability of the software packages and the characteristics of the datasets. Where applicable, it can be beneficial to use multiple methods and compare their results to make a more informed decision when determining the number of clusters. Additionally, incorporating domain knowledge and considering the practical utility of the resulting clusters is essential for ensuring that the clustering solution aligns with the goals and requirements of the analysis. Therefore, while there are no strict guidelines for choosing a method, a comprehensive approach that integrates various methods and takes into account domain expertise is typically the most effective strategy for determining the number of clusters.

4 Software packages

A summary of methods and R packages for clustering longitudinal is provided in Table 1. For model-based clustering, GBMT can be performed using *gbmt* (Magrini, 2022) and *flexmix* (Leisch, 2004) packages. LCMM can be performed using *lcmm* (Proust-Lima et al., 2017) and *mixAK* (Komárek & Komárková, 2014) packages. Of note, *mixAK* was originally developed to cluster multiple longitudinal features or outcomes, but it can also be used to cluster a single longitudinal feature. This package fits a model within the Bayesian framework using the MCMC algorithm, and therefore, prior knowledge may be incorporated. Furthermore, it assumes that the fixed effects are common between clusters whereas the random effects are different across clusters as described in Section 2.1.2.

For algorithm-based approaches, K-means clustering can be performed using the *kml* (Genolini & Falissard, 2010) and *kmlshape* (Genolini et al., 2016) packages, whereas hierarchical clustering can be performed using *clusterMLD* (Zhou et al., 2023) package.

For functional clustering approaches, K-centres clustering (Chiou & Li, 2007) can be performed using the *FClust*() function in the *fdapace* (Yao et al., 2005) package. However, this function requires the number of clusters to be known and does not provide a method to determine the number of clusters. Alternatively, one can use the *fdapace* (Yao et al., 2005) package to estimate the FPCA scores and subsequently apply a standard K-means clustering using *kmeans*() function of the *stats* package to obtain the clusters. This is the approach we used in the case studies (Section 5) and simulation study (Section 6) in the current paper. In addition, functional subspaces clustering can be performed using funHDDC (Bouveyron & Jacques, 2011) and funFEM (Bouveyron & Jacques, 2015) packages. Model-based clustering for functional data can be performed using *fdaMocca* (Pya Arnqvist et al., 2021) and *funcy* (Yassouridis et al., 2018) packages. Furthermore, the wavelet decomposition-based Gaussian model is implemented using *curviest* (Giacofci et al., 2013) package, and FADP is implemented using FADPclust (Ren et al., 2023) package.

Finally, we noted that a few R packages that were previously available have been removed from the Comprehensive R Archive Network (CRAN) repository (accessed on 2 February 2024). These packages include *kmlShape* (Genolini et al., 2016), *funcy* (Yassouridis et al., 2018), *funHDDC* (Bouveyron & Jacques, 2011) and *curvclust* (Giacofci et al., 2013). However, interested readers may implement these approaches using an earlier version of the package.

5 Case Studies

This section provides an empirical comparison among selected approaches using four datasets with distinct longitudinal structures. In particular, we considered four model-based approaches (*gbmt* (Magrini, 2022), *flexmix* (Leisch, 2004), *lcmm* (Proust-Lima et al., 2017), *mixAK* (Komárek & Komárková, 2014)), two algorithm-based approaches (*kml* (Genolini & Falissard, 2010), *clusterMLD* (Zhou et al., 2023)) and two functional clustering approaches (*fdapace* (Chen et al., 2017), *FADPclust* (Ren et al., 2023)).

5.1 ICTUS Cohort Study Data

The ICTUS cohort is a study of Alzheimer's disease (Reynish et al., 2007). From February 2003 to July 2005, 1380 patients were recruited from 29 specialist outpatient clinics of hospitals in 12 European countries. Most of these patients were seen during memory consultations and included consecutively. These patients were examined at 6-month intervals over 2 years. Each examination included a mini-mental score (MMS) assessment and a higher MMS indicates a better cognitive condition.

The data included in the current analysis contained 1374 patients. Each patient contributed to 15 measurements of MMS recorded at equally spaced (i.e. 6 months) intervals without missingness, leading to a total of 20 610 measurements. The time variable ranged from 1 to 15, indicating that the measurement recorded at the first interval to the fifteenth interval. This dataset corresponds to Scenario 1 in Figure 2. A longitudinal trajectory plot is shown in Figure S1, suggesting that there is substantial variation in individual development trajectory over time. It is critical to identify subgroups of patients that follow distinct trajectory patterns, which would provide evidence for targeted intervention to prevent further decline in their cognitive condition and facilitate predicting long-term outcomes of the patients.

To identify distinct trajectory patterns of MMS, we performed a cluster analysis based on the longitudinal cohort data. GBMT was performed using *gbmt* and *flexmix* packages. A quadratic function was used to model the trajectory pattern based on the observed data, and the number of clusters was determined by BIC. LCMM was performed using the *lcmm* and *mixAK* packages (Komárek & Komárková, 2014). For both packages, a quadratic function with random intercept and slopes (for both linear and quadratic terms) was used. In addition, for *lcmm*, we used the first-order autoregressive process to model the serial correlation. For *lcmm*, the number of clusters was determined by BIC while for *mixAK*, and the number of clusters was determined by PED. K-means clustering was performed using the *kml* package and hierarchical clustering using the *clusterMLD* package. The number of clusters was determined by the CH criterion for the former and by the gap statistics for the latter. For functional clustering, we performed K-means clustering on the FPCs estimated from the *fdapace* package, with the number of clusters determined by the CH criterion. Also, we performed FADP based on the distance metric (i.e. FADP1) using the *FADPclust* package. Unless otherwise specified, the default setting was used for each package. For each approach, we fitted the model for the number of clusters K ranging from 2 to 8, and the final model with the optimal number of clusters was determined based on the corresponding index. The time (in seconds) taken to run the final model was recorded. All approaches were run on a 12th Gen Intel(R) i9-12900 desktop computer.

The clustering results based on eight different approaches are shown in Figure 3. All approaches yielded distinct trajectory patterns suggesting the presence of heterogeneity of the MMS trajectories among these patients. However, the number of clusters, the proportion of individuals being assigned to each cluster and the trajectory patterns differed substantially among approaches. Specifically, GBTM based on *gbmt* identified eight clusters, three (Clusters 6, 7 and 8) of which showed rapid declining trends over time. GBTM based on *flexmix* identified five clusters, one (Cluster 5) of which showed rapid declining trends over time. LCMM based on *lcmm* identified six clusters, among which Cluster 5 showed a slow decline trend during the first 8 visits, but accelerated afterward, whereas Cluster 6 appeared to have a rapid decline during the first 8 visits but slowed down afterward. LCMM based on *mixAK* identified three clusters, among which Clusters 3 showed a rapidly declining trend over time. Algorithm-based and functional clustering all yielded a two-cluster model, with Cluster 1 showing constant high values of MMS, and Cluster 2 showing a slow decline trend over time.

For model-based approaches, the posterior cluster probabilities (defined in (6)) provide a measure of uncertainty in assigning individuals to specific clusters, with a higher probability suggesting a stronger belief in the assigned cluster, whereas a lower probability suggests greater uncertainty. It was found that all model-based approaches yielded a reasonably high posterior cluster probability for the resulting clusters (Figure S2). However, *gbmt* and *flexmix* appeared to have greater cluster certainty compared with *lcmm* and *mixAK*.

Next, we compared the agreement of cluster membership identified by different approaches using adjusted rand index (ARI) (Hubert & Arabie, 1985). A higher ARI value suggests a

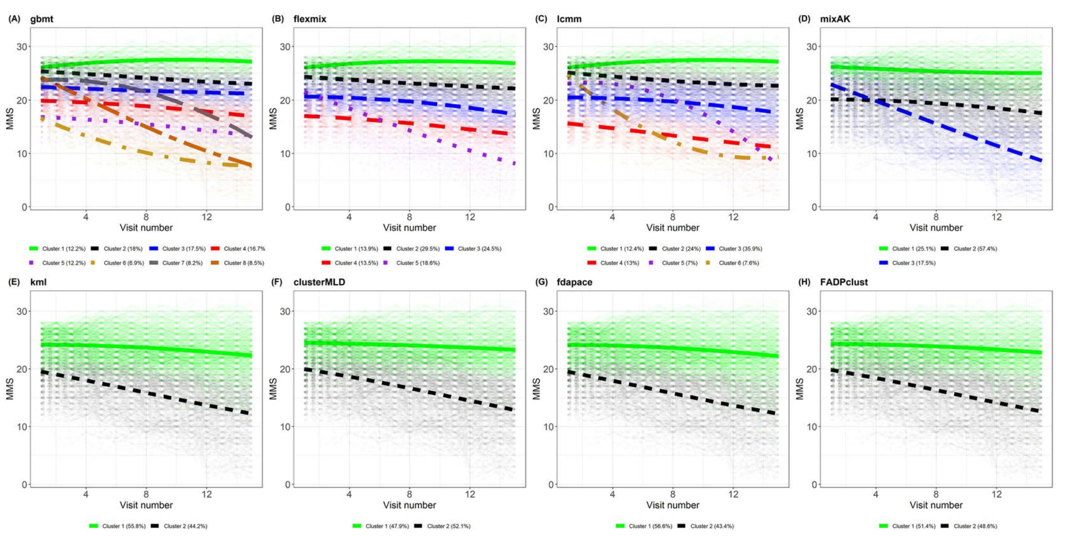

Figure 3. *Observed individual trajectories of mini-mental score with LOESS smoothing curves by clusters in the ICTUS cohort study. Clusters were derived from (A) GBTM based on gbmt package, (B) GBTM based on flexmix package, (C) LCMM based on lcmm package, (D) LCMM based on mixAK package, (E) K-means clustering based on kml package, (F) hierarchical clustering based on clusterMLD package, (G) functional clustering based on FPCA score estimated by fdapace package and K-means clustering, (H) functional clustering based on FADPclust package.*

greater agreement between the two approaches. It was shown that model-based approaches (*gbmt*, *flexmix*, *lcmm* and *mixAK*) yielded relatively low agreement (ARI < 0.6) between each other and between other approaches (Figure S3), while *kml* yielded high agreement with *clustMLD*, *fdapace* and *FADPclust*. The highest ARI was between *clustMLD* and *FADPclust*.

Computational speed also differed substantially between R packages (Figure S4). The majority of model-based approaches (with *flexmix* an exception) required a long computation time, such as *gbmt*, *lcmm* and *mixAK*, among which *lcmm* took the longest time compared with all other packages under study. In contrast, *flexmix*, *fdapace* and *kml* were among the fastest packages.

The current ICTUS cohort dataset does not contain other variables other than longitudinal MMS data. However, if additional information is available, it would be interesting to further characterise the resulting clusters and determine their utility by associating these clusters with baseline factors or future clinical outcomes.

5.2 Chronic Pelvic Pain Data

The Multi-Disciplinary Approach to the Study of Chronic Pelvic Pain (MAPP) (http://www.mappnetwork.org/) is a study aiming to help us better understand the pathophysiology of urologic chronic pelvic pain syndrome (UCPPS). This is a multifactorial constellation of symptoms characterised by one or more types and locations of urological pain, urinary urgency and/or urinary frequency symptoms. Currently, due to the lack of understanding of the aetiology and pathophysiology underlying the condition, treating patients with UCPPS is challenging, and no targeted treatment has been developed. Therefore, being able to identify patients' subgroups would provide insight into the changes in symptoms, which serves as an important foundation for discovering aetiologic and pathophysiological factors that are driving the pain and urinary symptoms.

In the MAPP study, UCPPS patients provided self-reported biweekly symptom data via web-based tools for 48 weeks. We focused on clustering the longitudinal urinary symptoms severity index (urinsev), a scale ranging from 0 to 25, where 0 suggests no urgency and 25 suggests the most severe urgency. This urinsev was derived as the sum of response (each 0–5) as the sum of responses to questions 5 and 6 (each 0–5) of the Genito Urinary Pain Index (GUPI) urinary subscale and response (each 0–5) to questions 1, 2 and 3 of the Interstitial Cystitis Symptom Index. In the current analysis, 397 patients were included, contributing to a total of 6974 observations. This dataset corresponds to Scenario 2 in Figure 2. A longitudinal trajectory plot of urinsev (Figure S5A) suggested that there was substantial variation in individual development trajectory over time, and the number of measurements differed substantially among patients (Figure S5B). The median (min, max) number of measurements was 20 (3, 25). The same dataset has been analysed and published previously by Guo et al. (2022).

To identify distinct trajectory patterns of urinsev, we performed a cluster analysis based on the longitudinal cohort data using the R packages in the first case study. However, *flexmix*, *kml* and *FADPclust* packages require the data to be complete (i.e. no missingness). Therefore, for these three approaches, we performed an imputation for missing data using a linear interpolation approach prior to the cluster analysis. For all other packages, we used the same specifications as before.

The clustering results based on eight different approaches are shown in Figure 4. GBTM based on *gbmt* and *flexmix* identified seven and six clusters, respectively. However, for *gbmt*, Clusters 1 and 2 did not well distinguish between each other (Figure 4A). LCMM based on *lcmm* and *mixAK* both identified three clusters; however, the pattern of Cluster 2 differed between the two packages. In particular, Cluster 2 in *lcmm* appeared to decrease rapidly during

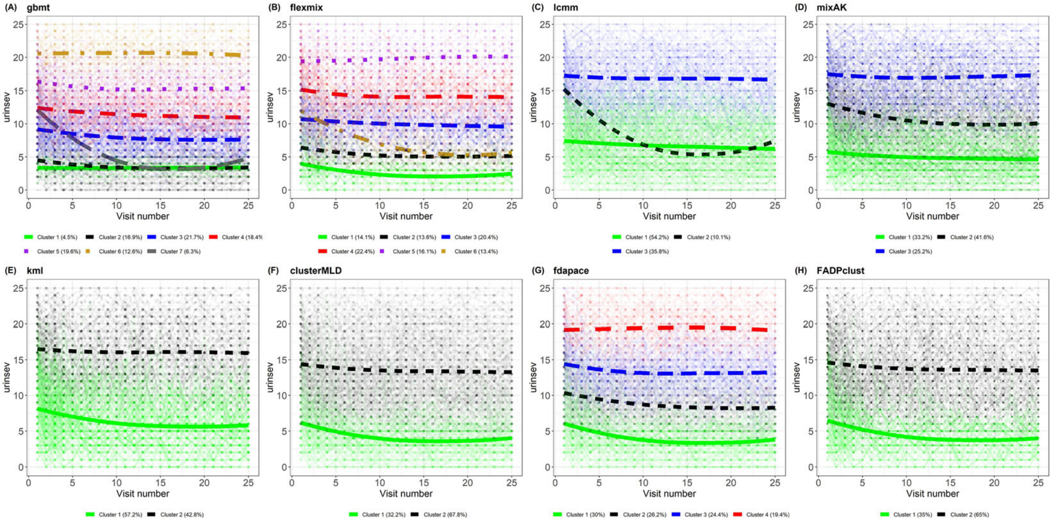

Figure 4. *Observed individual trajectories of urinary symptoms severity index with LOESS smoothing curves by clusters in the chronic pelvic pain data. Clusters were derived from (A) GBTM based on gbmt package, (B) GBTM based on flexmix package, (C) LCMM based on lcmm package, (D) LCMM based on mixAK package, (E) K-means clustering based on kml package, (F) hierarchical clustering based on clusterMLD package, (G) functional clustering based on FPCA score estimated by fdapace package and K-means clustering, (H) functional density peak clustering based on FADPclust package.*

the first 15 visits and then increase rapidly afterward. In contrast, Cluster 2 in *mixAK* declined slowly and remained steady after the 10th visit. *kml*, *clusterMLD* and *FADPclust* yielded two clusters, with Cluster 1 representing a low and stable urinary symptoms severity index and Cluster 2 representing a high and stable urinary symptoms severity index. On the other hand, *fdapace* with K-means clustering yielded four clusters representing four distinct levels of urinary symptoms severity index over time. Interestingly, the ARI suggested that agreement of cluster membership identified by different approaches remained relatively low (Figure S6), suggesting that the clustering solution, both in terms of the number of clusters and the cluster membership, obtained by different approaches could be inconsistent. However, the ARI between *clusterMLD* and *FADPclust* remained the highest. For each approach, the resulting clusters had distinct distribution in the RAND Interstitial Cystitis Epidemiology Study (RICE) painful urgency (presence vs. absence) (Figure S7). In this analysis, *lcmm* took the longest time followed by *mixAK* and *gbmt*, whereas *fdapace* took the shortest time followed by *flexmix* and *kml* (Figure S8).

5.3 Childhood Asthma Management Program Data

The Childhood Asthma Management Program (CAMP) is a randomised clinical trial of inhaled anti-inflammatory treatments for mild-to-moderate childhood asthma. It is a multicentre, masked, placebo-controlled, randomised trial. Between December 1993 and September 1995, a total of 1041 children aged 5–12 years were enrolled. Participants recruited to the trial were also followed by observational follow-up lasting 13 years. Both trial and observational follow-up included an annual prebronchodilator and postbronchodilator spirometry test as part of the protocol. The primary outcome of the trial was the lung function measured by the Forced Expiratory Volume at 1 s (FEV1). Higher FEV1 suggests better lung function. The trial found that the anti-inflammatory medications did not have a better long-term effect than placebo on lung function growth. Lung function has been associated with asthma in childhood and adulthood, we are interested in understanding the growth and decline in lung function in these patients with childhood asthma. These distinct lung function patterns could reveal links between asthma and subsequent chronic airflow obstruction, ultimately aiding in developing optimal personalised treatment strategies. This dataset has been analysed and published previously by McGeachie et al. (2016).

In this study, 1041 participants who had at least one FEV1 measurement were included in the analysis. These participants contributed to a total of 31 102 measurements. This dataset can be viewed as dense functional data with a structure corresponding to Scenario 4 in Figure 2. A longitudinal trajectory plot of FEV1 (Figure S9A) suggested that there was substantial variation in individual development trajectory over time, and the number of measurements differed substantially among patients (Figure S9B). The median (min, max) number of measurements was 33 (3, 37).

To identify distinct trajectory patterns of FEV1, we performed a cluster analysis based on the longitudinal cohort data using packages that can analyse unequally spaced and unbalanced longitudinal data. These packages included *gbmt*, *lcmm*, *mixAK*, *clusterMLD*, *fdapace*. For *gbmt*, a quadratic function was used, and for *lcmm* and*mixAK*, a quadratic function with random intercept and slopes (for both linear and quadratic terms) was used.

The clustering results based on eight different approaches are shown in Figure 5. For this dataset, *gbmt* failed to converge for $K \geq 2$, and therefore, we only reported the results for the rest of the approaches. Specifically, all packages except *mixAK* yielded two clusters with Cluster 1 (having a similar size as Cluster 2) representing individuals with better lung function (higher FEV1) compared with Cluster 2. The highest ARI was observed between *lcmm* and *fdapace*, followed by ARI between *clustMLD* and *fdapace* (Figure S10), and Cluster 2 was more likely to develop chronic obstructive pulmonary disease (COPD) at their last visit (Figure S11). In this analysis, *lcmm* took the longest whereas *clusterMD* took the least computation time (Figure S12).

5.4 Pregnant Women Hormone Data

The pregnant women hormone data come from a longitudinal study of young pregnant women who come to a private fertilisation obstetrics clinic in Santiago, Chile. During the first trimester of pregnancy at the prenatal examination, repeated measures of beta subunit human chorionic gonadotropin (β-HCG) were collected for each woman who participated in the study. The β-HCG is a type of hormone produced by the placenta after implantation. The current

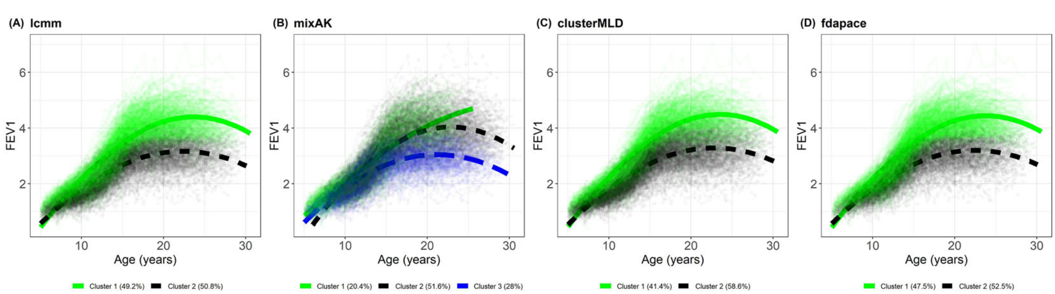

Figure 5. *Observed individual trajectories of FEV with LOESS smoothing curves by clusters in the CAMP data. Clusters were derived from (A) LCMM based on lcmm package, (B) LCMM based on mixAK package, (C) hierarchical clustering based on clusterMLD package, (D) functional clustering based on FPCA score estimated by fdapace package and K-means clustering.*

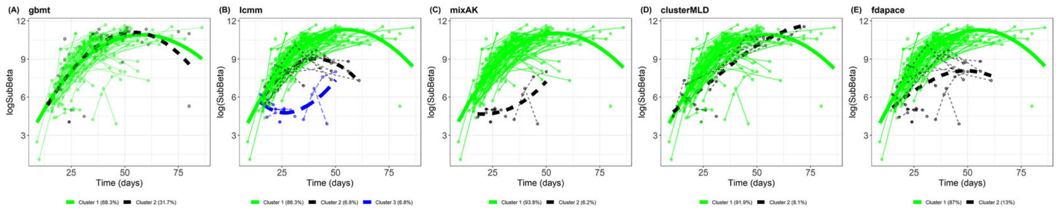

Figure 6. *Observed individual trajectories of β-HCG (in log scale) with LOESS smoothing curves by clusters in the pregnant women hormone data. Clusters were derived from (A) GBTM based on gbmt package, (B) LCMM based on lcmm package, (C) LCMM based on mixAK package, (D) hierarchical clustering based on clusterMLD package, (E) functional clustering based on FPCA score estimated by fdapace package and K-means clustering.*

analysis included 161 participants who had at least one β-HCG measurement collected at any given time during the study. The median (range) number of measurements per subject was 2 (1 to 6). By the end of follow-up, 124 were diagnosed with normal pregnancies (who had a terminal delivery) and 37 with abnormal pregnancies (who had spontaneous abortions or other types of adverse pregnancy outcomes). Identifying distinct β-HCG trajectories measured before the delivery is of great clinical interest which provides implications on the delivery status. This dataset can be viewed as very sparse data with a structure corresponding to Scenario 4 in Figure 2. Individual longitudinal trajectories of β-HCG (in nature log scale) and the distribution of the number of measurements are shown in Figure S13.

To identify distinct trajectory patterns of β-HCG, we performed a cluster analysis using packages with the same model specification as in the CAMP study. *gbmt*, *mixAK*, *clusterMLD* and *fdapace* yielded two clusters with Cluster 1 representing a majority of individuals who had a normal growth trajectory of β-HCG and Cluster 2 representing a small portion of individuals who had an abnormal growth trajectory (Figure 6). On the other hand, *lcmm* yielded three clusters, with Cluster 2 and Cluster 3, representing a small portion of individuals who deviated from the majority of individuals. Similar to the CAMP data, the highest ARI was observed between *lcmm* and *fdapace* (Figure S14), and an abnormal growth trajectory was associated with a higher proportion of abnormal pregnancies (Figure S15). In this analysis, *mixAK* took the longest time, followed by *lcmm* whereas *clusterMLD* took the least time (Figure S16).

6 Simulation Study

In this section, we perform simulation studies to compare the performance of these aforementioned approaches under three conditions. In Condition I (no random effect), repeated measurements within an individual were considered to be independent for a given cluster. In Condition II (common random effects), repeated measurements within an individual were considered to be correlated and the correlation (i.e. random effect distribution) was common between clusters. In Condition III (cluster-specific random effects), repeated measurements within an individual were considered to be correlated but the correlation was different between clusters (i.e. cluster-specific random effect distribution was specified).

6.1 Simulation Set-Up

We generated data from $K = 4$ clusters and the total sample size was set to be $N = 400$. The number of individuals for each cluster was set to be $N_k = 100$, where N_k denotes the number of individuals in cluster k, for $k = 1, 2, 3, 4$. The number of observations per individual is set to be

$n_i = 18$ (dense case) and $n_i = 6$ (sparse case) for all $i = 1, .., N$. In this simulation study, all individuals were set to have the same number of observations (corresponding to Scenarios 1 or 3 in Figure 2) such that all approaches can be applied and their performance can be compared.

The following three conditions were considered (Figure S17). For Condition I, we generated the data from $y_{ij} = f_{ik}(t_{ij}) + \epsilon_{ij}$, where $f_{ik}(t_{ij})$ is the mean trajectory for individual within cluster k. Specifically, we set $f_{i1}(t) = \cos(2\pi t)$, $f_{i2}(t) = -1 - 2\exp(-6t)$, $f_{i3}(t) = -2 - 2.5t$ and $f_{i4}(t) = 2.5 - 0.5t$. In this case, no random effects were specified, and therefore, individuals within the same cluster were assumed to have identical trajectories. We assumed the random error followed a normal distribution and was common to all clusters, that is, $\epsilon_{ij} \sim \mathcal{N}(0, 0.5^2)$. For Condition II, we generated the data from $y_{ij} = f_{ik}(t_{ij}) + r_i(t_{ij}) + \epsilon_{ij}$, where $r_i(t) = b_{i0} + b_{i1}t + b_{i2}t^2$ was the individual-specific random effect, where $\boldsymbol{b}_i = (b_{i0}, b_{i1}, b_{i2})^\top$ followed a multivariate normal distribution, that, $\boldsymbol{b}_i \sim N(0, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma} = \begin{pmatrix} 0.090 & 0.090 & -0.045 \\ 0.090 & 0.250 & -0.025 \\ -0.045 & -0.025 & 0.250 \end{pmatrix}$. In this case, the random effects were shared across all clusters. Therefore, individuals' trajectories within the same cluster were assumed to have variability and this variability was common for all clusters. Specification for $f_{ik}(t_{ij})$ and ϵ_{ij} remained the same as those in Condition I. For Condition III, we generated the data from $y_{ij} = f_{ik}(t_{ij}) + r_{ik}(t_{ij}) + \epsilon_{ij}$, where $r_{ik}(t) = b_{ik0} + b_{ik1}t + b_{ik2}t^2$ was individual-specific and cluster-specific random effect, where $\boldsymbol{b}_{ik} = (b_{ik0}, b_{ik1}, b_{ik2})^\top$ followed a multivariate normal distribution, that is, $\boldsymbol{b}_{ik} \sim N(0, \boldsymbol{\Sigma}_k)$, where $\boldsymbol{\Sigma}_1 = \begin{pmatrix} 0.160 & 0.144 & -0.072 \\ 0.144 & 0.360 & -0.036 \\ -0.072 & -0.036 & 0.360 \end{pmatrix}$, $\boldsymbol{\Sigma}_2 = \begin{pmatrix} 0.09 & 0.030 & -0.120 \\ 0.03 & 0.250 & 0.025 \\ -0.12 & 0.025 & 0.250 \end{pmatrix}$, $\boldsymbol{\Sigma}_3 = \begin{pmatrix} 0.040 & -0.048 & 0.024 \\ -0.048 & 0.160 & 0.016 \\ 0.024 & 0.016 & 0.160 \end{pmatrix}$, and $\boldsymbol{\Sigma}_4 = \begin{pmatrix} 0.010 & -0.004 & 0.016 \\ -0.004 & 0.040 & -0.004 \\ 0.016 & -0.004 & 0.040 \end{pmatrix}$. In this case, random effect distributions were dependent on the cluster membership k. Therefore, individuals' trajectories within the same cluster were assumed to have variability, but this variability could differ between clusters. Specification for $f_{ik}(t_{ij})$ and ϵ_{ij} remained the same as those in Condition I.

For all approaches, we ran the analysis from $K = 2$ to $K = 6$ and determined the optimal K based on the corresponding indices as used in the real data analyses. For model-based approaches, quadratic functions were used to capture the trajectory patterns. Also, in Condition I, for *lcmm* and *mixAK*, a random intercept was used, and therefore, in such a case, the two models were misspecified in terms of random effect distribution. In Conditions II and III, a random intercept and random slope were used, and therefore, in such cases, the two models were correctly specified in terms of random effect distribution. The simulation was repeated 100 times on a 12th Gen Intel(R) Core (TM) i7-12700 desktop computer. We reported the mean and standard deviation (SD) of the estimated number of clusters $\widehat{K}$, the ARI (Hubert & Arabie, 1985) under $\widehat{K}$ and under the true number of clusters K. The computational time in seconds was also recorded.

6.2 Simulation Results

Simulation results are summarised in Table 2. For Condition I (no random effect), model-based approaches using *gbmt* fully recovered the true number of clusters and the cluster membership, while *flexmix* slightly overestimated the number of clusters. *lcmm* performed reasonably well, yielding an estimated number of clusters very close to the true value and almost fully recovered the true cluster membership. *mixAK* underestimated the number of clusters leading to poor performance in recovering the individual cluster membership. However, given a true number of clusters, the *mixAK* yielded a reasonably well estimate of the cluster membership. The *gbmt*, *lcmm* and *mixAK* required longer computation time compared with all other approaches. For algorithm-based approaches, *clusterMLD* underestimated the number of clusters and did not fully recover the true number of cluster membership even if given the true number of clusters. *kml*, on the other hand, yielded a correct estimate of the number of clusters and the individual cluster membership. For FDA clustering approaches, *fdapace* slightly overestimated while *FADP* fully recovered the true number of clusters and the cluster membership. However, the former required the least computational time compared with all other approaches. Overall, the performance of these approaches was similar between dense and sparse cases.

For Condition II (common random effect), both *gbmt* and *flexmix* overestimated the number of clusters. The *lcmm* slightly overestimated the number of clusters but it performed reasonably well in recovering the true cluster membership, particularly when given the true number of clusters. *mixAK*, on the other hand, yielded an estimated number of clusters very close to the true value and performed well in recovering the cluster membership both under the estimated number of clusters and the true number of clusters. For algorithm-based approaches, *clusterMLD* underestimated the number of clusters and did not fully recover the true number of cluster membership even if given the true number of clusters, while *kml* yielded a correct estimate of the number of clusters and almost fully recovered the individual cluster membership. For FDA clustering approaches, both *fdapace* and *FADP* performed reasonably well, with the latter slightly outperforming the former both in estimating the number of clusters and the cluster membership. Still, *fdapace* took the least computational time compared with all other approaches. Similar results were observed for Condition III (cluster-specific random effect).

7 Discussion

In the current study, we provided a review of methods and R packages that could be used for clustering longitudinal data. Given the large number of literature, our review and comparison did not intend to be exhaustive; instead, we focused on commonly used approaches that could be implemented through available R packages. In addition, we made the connection between longitudinal data and functional data in the context of unsupervised learning and unlocked a set of functional clustering approaches for longitudinal data. We divided these approaches into three broad categories, namely, model-based, algorithm-based and functional clustering approaches. Of note, such categorisation is not mutually exclusive and it mainly helps to highlight the major differences between approaches. For example, within the category of functional clustering, approaches can be further categorised into model-based clustering and algorithm-based clustering (Jacques & Preda, 2014a; Zhang & Parnell, 2023). Careful consideration (e.g. based on subject-matter knowledge), evaluation of the model assumptions on a case-by-case basis, comparison between several possible approaches and investigation of the utility of the resulting clusterings are needed in practice.

Model-based clustering is often very flexible in handling different types of data, for example, sparse and irregularly measured longitudinal data. When observations are missing, model-based

Table 2. *Simulation results for comparing the performance of selected approaches*[1,2].

Methods	Dense ($n = 18$) $\widehat{K}$	$ARI\ (\widehat{K})$	$ARI\ (K)$	*Time (s)*	Sparse ($n = 6$) $\widehat{K}$	$ARI\ (\widehat{K})$	$ARI\ (K)$	*Time (s)*
Condition I (no random effect)								
gbmt	4 (0)	1 (0)	1 (0)	886.2 (860.13)	4 (0)	1 (0)	1 (0)	802.98 (620.27)
flexmix	4.65 (0.74)	0.88 (0.15)	0.87 (0.16)	1 (0.64)	4.26 (0.6)	0.94 (0.12)	0.94 (0.12)	0.95 (0.65)
lcmm	4.07 (0.26)	1 (0)	0.99 (0.04)	731.61 (266.13)	4.18 (0.5)	1 (0.03)	0.96 (0.1)	217.32 (145.98)
mixAK	3.47 (0.95)	0.77 (0.17)	0.92 (0.13)	145.11 (7.41)	3.33 (0.88)	0.75 (0.19)	0.91 (0.13)	377.08 (52.45)
clusterMLD	2.55 (0.87)	0.51 (0.28)	0.51 (0.28)	6.59 (0.07)	2.83 (0.89)	0.6 (0.25)	0.6 (0.25)	18.27 (3.05)
kml	4 (0)	1 (0)	1 (0)	7.87 (3.05)	4 (0)	1 (0)	1 (0)	3.75 (3.31)
fdapace	4.67 (0.78)	0.82 (0.19)	0.8 (0.21)	0.29 (0.35)	4.32 (0.47)	0.92 (0.13)	0.9 (0.17)	1.02 (0.57)
FADP	4 (0)	1 (0)	1 (0)	9.88 (1.18)	4 (0)	1 (0)	1 (0)	10.29 (1.37)
Condition II (common random effect)								
gbmt	6 (0)	0.78 (0.02)	0.94 (0.02)	341.4 (142.96)	6 (0)	0.78 (0.02)	0.94 (0.02)	309.59 (187.65)
flexmix	5.12 (0.81)	0.61 (0.05)	0.62 (0.06)	22.86 (9.88)	5.04 (0.83)	0.58 (0.06)	0.59 (0.05)	12.2 (3.42)
lcmm	4.87 (1.12)	0.89 (0.14)	0.99 (0.01)	1919.41 (1233.69)	4.92 (0.79)	0.92 (0.07)	0.96 (0.07)	705.42 (204.42)
mixAK	3.9 (0.39)	0.96 (0.11)	0.97 (0.09)	519.68 (115.2)	3.79 (0.61)	0.89 (0.17)	0.95 (0.1)	403.63 (56.12)
clusterMLD	2.36 (0.7)	0.45 (0.22)	0.45 (0.22)	19.73 (2.92)	2.71 (0.77)	0.57 (0.23)	0.57 (0.23)	15 (6.7)
kml	4 (0)	0.95 (0.02)	0.95 (0.02)	14.62 (7.35)	4 (0)	0.94 (0.02)	0.94 (0.02)	13.88 (5.86)
fdapace	4.06 (0.24)	0.92 (0.07)	0.91 (0.08)	1.27 (1.03)	4.01 (0.1)	0.88 (0.05)	0.88 (0.05)	1.34 (1.02)
FADP	4 (0)	0.97 (0.02)	0.97 (0.02)	13.52 (2.64)	4 (0)	0.96 (0.02)	0.96 (0.02)	9.62 (5.96)
Condition III (cluster-specific random effect)								
gbmt	6 (0)	0.83 (0.03)	0.98 (0.02)	69.21 (29.09)	5.99 (0.1)	0.82 (0.03)	0.97 (0.01)	29.91 (16.92)
flexmix	4.66 (0.73)	0.64 (0.06)	0.64 (0.07)	16.04 (3.67)	5.1 (0.72)	0.6 (0.06)	0.6 (0.05)	7.43 (1.39)
lcmm	5.53 (0.58)	0.88 (0.05)	1 (0.01)	410.74 (100.09)	5.73 (0.45)	0.86 (0.04)	0.98 (0.01)	121.29 (48.22)
mixAK	3.76 (0.62)	0.91 (0.17)	0.98 (0.08)	263.36 (19.41)	3.86 (0.4)	0.93 (0.12)	0.97 (0.08)	204.25 (12.12)
clusterMLD	2.2 (0.71)	0.38 (0.16)	0.38 (0.16)	19.65 (3.21)	2.33 (0.75)	0.42 (0.2)	0.42 (0.2)	18.26 (3.16)
kml	4 (0)	0.96 (0.02)	0.96 (0.02)	10.59 (7.83)	4 (0)	0.95 (0.02)	0.95 (0.02)	9.72 (6.68)
fdapace	4.36 (0.59)	0.92 (0.05)	0.95 (0.02)	0.58 (0.46)	4.16 (0.42)	0.9 (0.04)	0.91 (0.03)	0.94 (0.42)
FADP	3.99 (0.1)	0.98 (0.04)	0.98 (0.03)	8.62 (1.11)	4 (0.25)	0.97 (0.06)	0.96 (0.07)	3.12 (0.1)

[1]gbmt: Group-based trajectory model using *gbmt* package; flexmix: Group-based trajectory model using *flexmix* package; lcmm: latent class mixed effect model using *lcmm* package; mixAK: latent class mixed effect model using *mixAK* package; clusterMLD: Hierarchical clustering based on *clusterMLD* package; kml: K-means clustering using *kml* package; fdapace: functional clustering based on FPCA score estimated by *fdapace* package and K-means clustering; FADPclust: functional clustering based on *FADPclust* package.

[2]$\widehat{K}$: estimated number of clusters; ARI ($\widehat{K}$): adjusted rand index under the estimated number of clusters; ARI (K): adjusted rand index under the ture number of clusters.

clustering methods rely on full-information maximum likelihood (FIML) estimation to integrate all available information based on missing at random (MAR) (i.e. ignorable) assumption (Little & Rubin, 2014). Although the automated FIML option is attractive, the efficiency gain of any missing data technique will be constrained by the amount and the type of missing data within a particular dataset. In addition, this assumption of MAR may not hold under certain scenarios and needs to be relaxed. For example, for Scenarios 2 and 4 in Figure 2, it is likely that the timing and/or frequency of the measurements are associated with the disease course and therefore inform their cluster membership. In such a case, the visiting process can be considered as a random process and a sub-model can be employed to model this process (Pullenayegum & Lim, 2016). Both non-ignorable intermittent missing (Lin et al., 2004) or non-ignorable dropout (Muthén et al., 2011) can be considered in future studies. Another feature of the model-based approach is that it yields a posterior probability of cluster membership, which quantifies the uncertainty of the cluster assignment and serves as a measure of the quality of the clustering solution.

In particular, GBTM and LCMM are two commonly used model-based approaches. Both models require the data (i.e. error distribution) to come from an underlying normal distribution. The latter further requires the random effects to come from a normal distribution. When these assumptions are violated, mixture models with non-normal error distribution (Depaoli et al., 2019) and/or non-normal random effect distributions (Muthén & Asparouhov, 2015) can be considered as alternatives. These models may arise from the structural equation modelling framework and therefore can be implemented using the Mplus Program (Muthén & Muthén, 2017). Also, the GBTM assumes that individual trajectories have identical patterns within a given cluster and can be viewed as a special case of LCMM, the latter of which allows individual trajectories to deviate from their mean trajectory within the cluster. Because of this assumption, a large number of clusters are needed to account for the population heterogeneity in GBTM as compared with LCMM, as seen in our case studies. However, compared with LCMM, GBTM is generally faster and may be useful when a more complex model such as LCMM fails to converge or takes much longer computation time. Typically, polynomial terms are used to model the trajectory patterns in model-based clustering. While this is flexible and interpretable, it can be suboptimal for the complex non-linear relationship.

Algorithm-based approaches serve as a good alternative when one believes that the assumptions for model-based approaches are not met. In particular, K-means clustering (Genolini & Falissard, 2010) relies on distance metrics to quantify cluster dissimilarity and does not make assumptions about the shape of the trajectory, and the hierarchical clustering (Zhou et al., 2023) uses a B-spline basis to approximate the trajectory pattern over time (which is viewed as functional data), and therefore, both approaches are more flexible when handling non-linear longitudinal data. However, the current version of *kml* can not handle sparse and irregular longitudinal data, and missing data must be imputed before the cluster analysis, whereas *clusterMLD* is more flexible and is appliable to sparse and irregular longitudinal, and data with missingness assuming MAR.

Functional clustering approaches view observations coming from infinite-dimensional space (i.e. functions). These functions are often represented as curves, and the goal is to analyse the properties of these functions, such as their smoothness or variability, rather than to estimate specific parameters of a probability distribution. Dimension reduction techniques such as spline basis and FPC scores are commonly used to approximate the curves prior to performing cluster analysis. Therefore, a new set of analytical tools is developed for FDA, which can be more flexible and robust in handling complex and diverse data sets. This flexibility allows it to be used in a wide variety of applications where the data may have unknown or complex distributions, a large number of observations (i.e. high dimensionality) and non-linear relationships. The functional clustering approaches considered in the case studies and simulation study, that is, *fdapace* and *FADPclust*, appear to perform reasonably well in discovering distinct longitudinal trajectory patterns and in general computationally efficient.

In our real data applications, it was found that model-based methods can generally identify more clusters, and some clusters had quite different slopes from others. Meanwhile, the algorithm-based and FDA-based methods generally identified clusters with parallel trajectories (e.g. high vs. low) but cannot identify trajectories with rapid changes as a separate cluster (e.g. Figures 3 and 4). This is because model-based methods typically involve fitting parametric models to the data. By allowing each cluster to have its own parameters (such as intercepts and slopes), these models have the flexibility to capture heterogeneity in trajectories with varying rapid change patterns. Algorithm-based methods, such as K-means clustering, rely on distance-based measures to group similar data points together. These methods may struggle to capture clusters with different slopes or rapid changes because they do not explicitly model trajectory shapes or changes over time unless a special distance metric is used. Instead, they tend to identify clusters based on overall similarities in the data, which may result in clusters with parallel trajectories. FDA approaches are non-parametric approaches and involve representing longitudinal data as smooth functions and then applying statistical techniques to analyse these functions. While FDA methods can capture overall complex trends and shapes of trajectories effectively, they may not be as effective in identifying clusters with rapid changes or distinct slopes compared with parametric approaches. The quality of clustering can be further assessed by measures of cluster stability in practice, for example, using subsetting, jittering the data points and replacing some data points with artificial noise points (Hennig, 2007).

The underlying assumption of the methods reviewed in our study is that there exist clusters (i.e. subpopulations) and that the individual cluster membership does not change over time (i.e. time-independent). However, such assumptions may not be true in some practical applications. In such a case, dynamic clustering may be considered in which the individual cluster membership is allowed to be updated once new observations are available. Example of such models includes dynamic latent class analysis (Lin et al., 2014) and mixed hidden Markov model (Raffa & Dubin, 2015). Moreover, the current review focused on methods for clustering a single longitudinal feature. However, methods for clustering multiple features have been emerging and gaining increasing popularity. For example, these methods include group-based multi-trajectory modelling (Nagin et al., 2018), growth curve mixture model to jointly model multiple longitudinal outcomes (Frühwirth-Schnatter & Pyne, 2010; Lu & Lou, 2022; Neelon et al., 2011; Tan et al., 2022) and algorithm-based approaches (Genolini et al., 2013). In addition, software packages such as *gbmt*, *flexmix*, *lcmm*, *mixAK* and *clusterMLD* have the functionality that can be used for clustering multiple longitudinal features. The readers can refer to a recent study for a comparison of these methods and packages for multiple longitudinal features (Lu, Ahmadiankalati, & Tan, 2023), in which a single longitudinal feature can be considered a special case. Other advanced clustering approaches, such as for multi-source longitudinal data (Lu, Subbarao, & Lou, 2023) and high-dimensional longitudinal data (Lu & Chandra, 2024) have been emerging and will be of interest to researchers seeking a comprehensive understanding of complex longitudinal data structures and their underlying patterns. While our review focuses on continuous longitudinal data, methods for clustering other types of longitudinal data have also been developed, for example, categorical data (Lanza & Cooper, 2016), count data (Subtil et al., 2017), zero-inflated data (Nielsen et al., 2014) and ordinal data (Costilla et al., 2019). In particular, some of the R packages discussed here can also be used for other types of data. For example, both *flexmix* and *mixAK* can be applied to binary and count data, while *lcmm* can be applied to binary and ordinal data. Apart from R, other software packages may also be used, such as SAS, Mplus and Stata. Readers may refer to van der Nest et al. (2020) for a discussion of different software.

In practice, it is often of interest to relate the individual cluster membership to predictors (e.g. baseline factors) if such information is available. Such analyses could further demonstrate the utility of the clusters in reflecting the disease conditions. Compared with the algorithm-based and FDA approaches, the model-based approach provides a more flexible framework to account for additional predictors (Table 1). When covariate effects on predicting the cluster membership are of interest, there are two possible approaches to achieve this goal (Kim et al., 2016). The first and perhaps the most common approach is the stepwise approach, also known as the three-step approach. This analysis usually consists of three steps: (a) identifying the clusters without including additional predictors; (b) participants' posterior membership probabilities from the first step are used to create the most likely cluster membership; (c) the cluster membership variable derived from the model is used as the outcome variable and regressed on the predictors while adjusting for the uncertainty in the cluster assignments (Vermunt, 2010). With this approach, the predictors do not affect the clustering solution, avoiding the scenarios in which including different predictors could lead to a different clustering, leading to difficulty in results interpretation. In addition, no explicit assumptions about the predictors are made. Clustering and regression analysis are two separate steps that may facilitate the result interpretation. Examples of the stepwise approach can be found in the literature (Rzehak et al., 2013; Stull et al., 2011). An alternative approach is joint modelling, also known as the one-step approach. With this approach, clusters are estimated jointly with their predictors in one overall model, in which the predictors contribute to the clustering solution. This allows the full model to be estimated and most data information to be incorporated. However, the predictors could influence the cluster solution, leading to different clustering for different sets of predictors and difficulty in result interpretation. If the model is correctly specified, this approach may improve the clustering and reduce the standard errors of the predictor effects (Clark & Muthén, 2009).

In summary, clustering longitudinal data is a complicated analysis process. The current study provides a review of commonly used methods when clustering longitudinal data is a research goal. Appropriately assigning individuals into clusters that are practically meaningful requires subject-specific knowledge and appropriate methods that are able to capture the difference between trajectories. The choice of one set of approaches versus the others depends on the research questions of interest, assumptions that can be made and the data types at hand.

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Conflict of Interest

None declared.

Data Availability Statement

The ICTUS cohort study data can be found in Genolini et al. (2016). The chronic pelvic pain data can be found in Guo et al. (2022). The CAMP data can be found in McGeachie et al. (2016). The pregnant women hormone data are not publicly available.

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